



DOMINIK SZEWCZYK

DATA SCIENTIST

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EDUCATION

2022 – 2026

BREDA UNIVERSITY OF
APPLIED SCIENCES

- Data Science & Artificial Intelligence

2019 – 2023

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MODRZEWSKIEGO

- Extended Mathematics Profile
- Top-ranked school in Silesia

SOFT SKILLS

- Reliability
- Problem-solving
- Adaptability
- Time management
- Teamwork
- Communication

LANGUAGES

- English (Fluent)
- Polish (Fluent)
- German (Basics)

PROFILE

Ambitious Data Science & AI student with a strong track record of solving real-world problems through machine learning, computer vision, and NLP. I've delivered impactful solutions in logistics, plant science, and media analytics, often building end-to-end systems including dashboards, predictive models, and cloud-based deployments. I enjoy working on interdisciplinary teams and delivering practical, measurable results.

TECHNICAL SKILLS

Core: Machine Learning, Deep Learning, NLP, Computer Vision, Data Analysis, Predictive Modeling

Tools & Languages: Python, Keras, scikit-learn, TensorFlow, Streamlit, Power BI, FastAPI, Docker, Git

Cloud & MLOps: Microsoft Azure, Azure ML, CI/CD Pipelines

Other: Data Visualization, Feature Engineering, Agile (Scrum)

WORK EXPERIENCE

Data Science Intern

Feb 2025 - Jun 2025

De Rooy Transport & Logistics

Developed an intelligent system to forecast unexpected pickups and reduce unnecessary driving across the Netherlands, resulting in measurable savings in cost, emissions, and planning time.

- Developed a multi-model ML pipeline (Keras, RandomForest, XGBoost) to predict pickup occurrence, location, size, and insertion time
- Engineered spatial, temporal, and calendar features to improve accuracy
- Created an interactive Streamlit dashboard to upload data, train models, and visualize predictions
- Enabled up to 1500 km driving reduction and 900 kg CO₂ savings per day

HOBBY

Since I can remember, I have actively participated in various sports, including football and gym, as well as chess, where I hold a 3rd category. Last year, I completed a half marathon in 1 hour and 43 minutes.

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SELECTED PROJECTS

**Engineer MLOps:
Deploying AI Models
for Plant Science**

*Netherlands Plant
Eco-phenotyping
Center (NPEC)*

- Transformed ML models into production-ready solutions for the ATHENA application
- Designed and deployed automated pipelines using Microsoft Azure, Docker, and FastAPI
- Applied computer vision techniques for organ segmentation and landmark detection, boosting phenotyping accuracy and research efficiency
- Received public recognition from NPEC for advancing AI-based image analysis

**Emotion Classification
System for Video
Content**

Banijay Media

- Developed a pipeline for emotion detection in video content using NLP and transformer models (BERT)
- Achieved an F1 score above 0.88 on the Kaggle test set through optimized model selection and fine-tuning
- Evaluated and compared models including LSTM and CRF to enhance classification accuracy from extracted text

**AI & Robotics Pipeline
for Root Segmentation
and Precision Handling**

*Netherlands Plant Eco-
phenotyping Center
(NPEC)*

- Built a U-Net model from scratch for plant root segmentation, achieving a SAMP score under 13
- Applied instance segmentation to detect root landmarks with 1mm accuracy, improving data quality for phenotyping
- Simulated a PID-controlled liquid handling robot using computer vision for precise delivery to root tips

**AI Insights for Facility
Management Studies**

*Breda University of
Applied Sciences*

- Conducted a mixed-methods study using survey design, data collection, and statistical analysis in R
- Delivered actionable insights for facility management through AI-driven analysis
- Built an interactive Streamlit dashboard showcased at the AI at BUAS Conference to support stakeholder decisions

**Audience Preferences
Analysis for Banijay**

Banijay Media

- Collaborated with Banijay Media to analyze content and ratings data, linking insights on audience preferences and social media impact to enhance their content strategy.
- Developed a predictive model to forecast ratings, providing actionable insights that aimed to support data-driven decision-making.

**Renewable Energy
Dashboard for SDG
Hub**

*Presented at SDG
Conference*

- Built an interactive Power BI dashboard to analyze European renewable energy trends aligned with SDG 7
- Performed exploratory data analysis using SDGDataBank and applied linear regression to model consumption trends
- Presented findings at a conference; SDG Hub praised the dashboard as highly informative and intuitive